

# The sharp universal constant for Gaussian entropic-transport coupling convergence

Candidate solution packet for arXiv:2512.19457

21 July 2026

**Status: full solution, likely valid (human review requested).** For every dimension  $d$ , the sharp constant in Conjecture 3.1 of Yun is  $C_d = d/2$ . In fact, the normalized squared Wasserstein distance has an explicit limit for every positive-definite covariance pair, and equality in the universal bound occurs exactly for identity transport  $A = B$ .

## 1 Source and open problem

The source is Ho Yun's arXiv:2512.19457 paper [1]. Its Conjecture 3.1 appears on page 19 of the current PDF. The paper considers the entropic Gaussian coupling  $\pi_\varepsilon$  between  $N(0, A)$  and  $N(0, B)$  and its unregularized limit  $\pi_0$ .

Spectral Shrinkage of Gaussian Entropic Optimal Transport

**Conjecture 3.1** (Universal Bounds). *While the asymptotic scaling  $\mathcal{W}_2^2(\pi_\varepsilon, \pi_0) = \mathcal{O}(\varepsilon)$  is guaranteed in the non-degenerate finite-dimensional setting  $(\mathbf{A}, \mathbf{B} \succ \mathbf{0})$  on  $\mathcal{H} = \mathbb{R}^d$  from (11), determining the sharp universal constant governing this rate remains an open problem. Specifically, does there exist a dimension-dependent constant  $C_d > 0$  such that*

$$\limsup_{\varepsilon \downarrow 0} \frac{\mathcal{W}_2^2(\pi_\varepsilon, \pi_0)}{\varepsilon} \leq C_d \quad \text{for all } \mathbf{A}, \mathbf{B} \succ \mathbf{0}?$$

*Analytical results for the identity transport case  $(\mathbf{A} = \mathbf{B})$  in (10) gives a lower bound of  $C_d \geq d/2$ . However, our numerical experiments with randomly generated full-rank covariances (see Fig. 3) consistently exhibit a strictly smaller rate, typically concentrating around  $\approx 0.36d$ . It remains unclear whether the identity case represents a theoretical worst-case scenario, or if the observed lower rates are simply an artifact of the concentration of measure in random matrix ensembles.*

Figure 1: Conjecture 3.1 in the source paper, page 19.

**Transcription.** For  $A, B \succ 0$  on  $\mathbb{R}^d$ , is there a dimension-dependent constant  $C_d > 0$  such that

$$\limsup_{\varepsilon \downarrow 0} \frac{\mathcal{W}_2^2(\pi_\varepsilon, \pi_0)}{\varepsilon} \leq C_d \quad \text{for all } A, B \succ 0?$$

The source observes that  $A = B$  gives the lower bound  $C_d \geq d/2$  and asks whether this identity case is the worst case.

## 2 Notation and the source formula

All matrices below are real symmetric unless stated otherwise. Let

$$P = A^{1/2} B A^{1/2}, \quad Z = P^{1/2}, \quad X = A, \quad Y = Z A^{-1} Z, \quad S = X + Y.$$

Because  $A, B \succ 0$ , all five matrices  $P, Z, X, Y, S$  are positive definite. With the properly aligned Green operators

$$G = A^{1/2}, \quad M = A^{-1/2}Z,$$

one has  $GG^* = A$ ,  $MM^* = B$ ,  $G^*M = Z$ ,  $G^*G = X$ , and  $M^*M = Y$ . The spectral shrinkage is

$$R_\varepsilon = f_\varepsilon(Z), \quad f_\varepsilon(t) = \frac{2t}{\sqrt{4t^2 + \varepsilon^2} + \varepsilon}.$$

Theorem 3.10 of the source gives

$$\mathcal{W}_2^2(\pi_\varepsilon, \pi_0) = 2 \operatorname{tr} \left( S - Q_\varepsilon^{1/2} \right), \quad (1)$$

$$Q_\varepsilon = X^2 + Y^2 + XR_\varepsilon Y + YR_\varepsilon X. \quad (2)$$

Here  $\operatorname{tr} S = \operatorname{tr}(A + B)$ , since  $Y = M^*M$  and  $MM^* = B$ .

### 3 Proof intuition

The exact formula (1) only needs to be differentiated at  $\varepsilon = 0$ . Spectral shrinkage has first variation  $R_\varepsilon = I - (\varepsilon/2)Z^{-1} + O(\varepsilon^2)$ . The apparently noncommutative cross terms in (2) then collapse because

$$XZ^{-1}Y = YZ^{-1}X = Z.$$

Thus  $Q_\varepsilon = S^2 - \varepsilon Z + O(\varepsilon^2)$ , and differentiating the trace of the matrix square root produces the exact asymptotic coefficient  $\operatorname{tr}(S^{-1}Z)$ .

The bound is an operator arithmetic–geometric mean identity rather than a spectral estimate:

$$S - 2Z = X + ZX^{-1}Z - 2Z = (X - Z)X^{-1}(X - Z) \succeq 0.$$

After congruence and inversion, this gives  $Z^{1/2}S^{-1}Z^{1/2} \preceq \frac{1}{2}I$ . Taking the trace yields  $d/2$ .

### 4 Sharp theorem

**Theorem 1.** *Let  $A, B \in \mathbb{R}^{d \times d}$  be positive definite, and let  $\pi_\varepsilon, \pi_0$  be the Gaussian couplings in Conjecture 3.1 of [1]. With  $Z = (A^{1/2}BA^{1/2})^{1/2}$ , the limit exists and is*

$$\lim_{\varepsilon \downarrow 0} \frac{\mathcal{W}_2^2(\pi_\varepsilon, \pi_0)}{\varepsilon} = \operatorname{tr} \left[ (A + ZA^{-1}Z)^{-1}Z \right] \leq \frac{d}{2}. \quad (3)$$

Equality holds in (3) if and only if  $A = B$ . Consequently the smallest universal constant in Conjecture 3.1 is

$$\boxed{C_d = d/2}.$$

**Lemma 1** (Trace-square-root differential). *If  $C \succ 0$  and  $H = H^*$ , then, as  $t \rightarrow 0$ ,*

$$\operatorname{tr}(C + tH)^{1/2} = \operatorname{tr} C^{1/2} + \frac{t}{2} \operatorname{tr}(C^{-1/2}H) + o(t).$$

*The same conclusion holds when  $tH$  is replaced by  $tH + O(t^2)$ .*

*Proof.* Diagonalize  $C$ . The standard divided-difference formula for the Fréchet derivative of a matrix function says that the diagonal entries of  $D(\sqrt{\cdot})_C[H]$  in this eigenbasis are  $\frac{1}{2}\lambda_i^{-1/2}H_{ii}$ . Their sum is  $\frac{1}{2} \operatorname{tr}(C^{-1/2}H)$ . Since the square-root map is continuously Fréchet differentiable on the positive-definite cone, an  $O(t^2)$  remainder in the argument contributes  $o(t)$ . This is also the usual spectral trace differentiation identity; see, for example, Higham [2, Chapter 3].  $\square$

*Proof of Theorem 1.* On the positive spectrum of  $Z$ , scalar Taylor expansion, uniformly over that finite spectrum, gives

$$f_\varepsilon(Z) = I - \frac{\varepsilon}{2}Z^{-1} + O(\varepsilon^2) \quad (\varepsilon \downarrow 0)$$

in operator norm. Since  $Y = ZX^{-1}Z$ ,

$$XZ^{-1}Y = Z, \quad YZ^{-1}X = Z.$$

Substitution in (2) yields

$$\begin{aligned} Q_\varepsilon &= X^2 + Y^2 + XY + YX - \frac{\varepsilon}{2}(XZ^{-1}Y + YZ^{-1}X) + O(\varepsilon^2) \\ &= S^2 - \varepsilon Z + O(\varepsilon^2). \end{aligned}$$

Apply Lemma 1 with  $C = S^2$  and  $H = -Z$ :

$$\mathrm{tr} Q_\varepsilon^{1/2} = \mathrm{tr} S - \frac{\varepsilon}{2} \mathrm{tr}(S^{-1}Z) + o(\varepsilon).$$

Equation (1) now proves the equality in (3).

It remains to bound this coefficient. Using  $Y = ZX^{-1}Z$ ,

$$S - 2Z = X + ZX^{-1}Z - 2Z = (X - Z)X^{-1}(X - Z) \succeq 0. \quad (4)$$

Congruence by  $Z^{-1/2}$  and inversion give

$$Z^{1/2}S^{-1}Z^{1/2} \preceq \frac{1}{2}I.$$

Therefore

$$\mathrm{tr}(S^{-1}Z) = \mathrm{tr}\left(Z^{1/2}S^{-1}Z^{1/2}\right) \leq \frac{d}{2}.$$

If equality holds, the positive-semidefinite matrix  $\frac{1}{2}I - Z^{1/2}S^{-1}Z^{1/2}$  has trace zero and hence vanishes. Thus  $S = 2Z$ . Equation (4) then forces  $X = Z$ . Since  $Z^2 = A^{1/2}BA^{1/2}$  and  $X = A$ , this is equivalent to  $A^{1/2}BA^{1/2} = A^2$ , hence  $B = A$ . Conversely,  $A = B$  gives  $Z = A$  and  $\mathrm{tr}(S^{-1}Z) = \mathrm{tr}((2A)^{-1}A) = d/2$ . The source already identifies this case as the lower-bound example, so  $C_d = d/2$  is sharp.  $\square$

## 5 Verification and limitations

- The proof uses only the source’s exact Wasserstein formula (1), finite-dimensional matrix Taylor expansion, and the explicit positive-semidefinite identity (4).
- The accompanying script tested five independent random covariance pairs in each of dimensions 1, 2, 5, 10. For  $\varepsilon \in \{10^{-3}, 3 \cdot 10^{-4}, 10^{-4}\}$ , direct evaluation of the source formula converged to (3); every coefficient was at most  $d/2$ . It also checks equality for  $A = B$ . These tests are sanity checks, not part of the proof.
- Scope: the theorem answers exactly the source’s non-degenerate, finite-dimensional conjecture. It makes no new claim about singular covariances or infinite-dimensional nonparametric rates.

**Bounded novelty check.** On 21 July 2026, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2512.19457 and the phrases “spectral shrinkage”, “Gaussian entropic optimal transport”, and “universal bounds”. No prior packet or duplicate was found. A web/arXiv search used the exact paper title with “Universal Bounds”, “Wasserstein Gaussian entropic coupling  $d/2$ ”, “Gaussian entropic optimal transport coupling convergence sharp constant”, and a site-restricted arXiv variant. The current arXiv source still states Conjecture 3.1, and no separate primary source resolving this exact constant was found. This is a bounded search, not a proof of novelty.

**Human-review recommendation.** Verify the transcription of Theorem 3.10 into (1), including the normalization of  $\varepsilon$ , and the aligned factor  $M = A^{-1/2}Z$ . Once those convention-sensitive inputs are confirmed, the remaining expansion and operator inequality are self-contained.

## References

- [1] H. Yun, *Spectral Shrinkage of Gaussian Entropic Optimal Transport*, arXiv:2512.19457 (2025; current PDF accessed 21 July 2026), <https://arxiv.org/abs/2512.19457>.
- [2] N. J. Higham, *Functions of Matrices: Theory and Computation*, Society for Industrial and Applied Mathematics, 2008.